



POLITECNICO
MILANO 1863

Dynamics of Electrical Machines and Drives

Induction Motor model

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Induction Motor Modeling

Approach

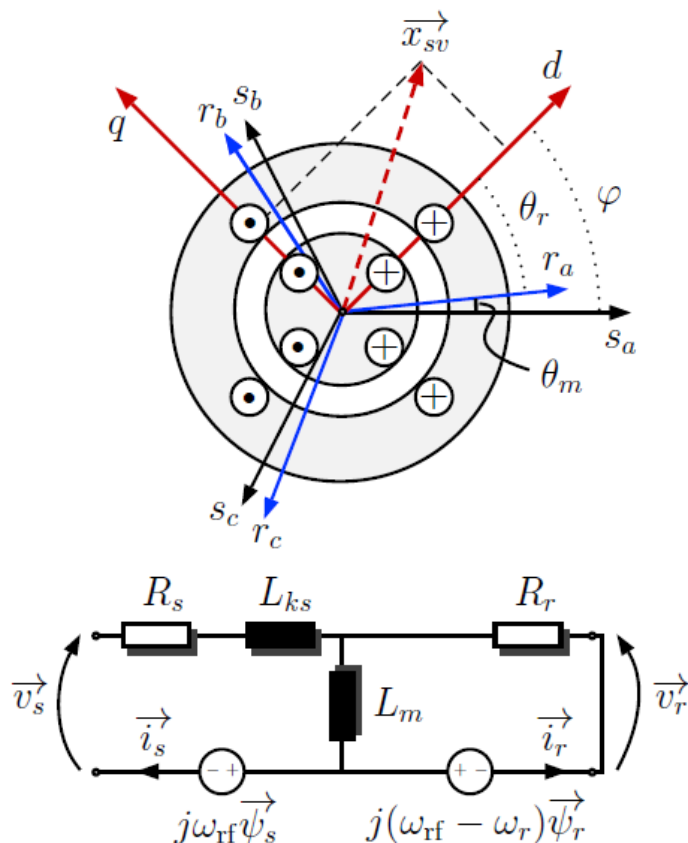
- The **modeling process** of induction machines is typically done in two/three steps:
 - ① starting from three-phase *abc* quantities and fundamental physical laws, such as Faraday's law of induction and Lorentz force, we derive machine's differential equations
 - ② transform the model into an orthogonal reference frame to simplify the representation, here we can use
 - **space vector notation**
 - **matrix notation**
 - state-space notation
 - ③ sometimes the machine model is translated from SI to normalized quantities (e.g. per unit)

(The differential equations derivation is beyond the scope of this lesson)

Model in Space Vector Notation

Induction Motor

- The dynamic model of a three-phase IM with *orthogonal* axes dq is:



$$\Gamma\text{-model} \rightarrow L_{ks} = L_s + L_m^2/L_r$$

IM motor model (sv form)

$$\vec{v}_s = R_s \vec{i}_s + p\vec{\psi}_s + j\omega_{rf}\vec{\psi}_s$$

$$\vec{v}_r = R_r \vec{i}_r + p\vec{\psi}_r + j(\omega_{rf} - \omega_r)\vec{\psi}_r$$

$$\vec{\psi}_s = L_{ks} \vec{i}_s + \vec{\psi}_r$$

$$\vec{\psi}_r = L_m(\vec{i}_s + \vec{i}_r)$$

$$m_e = n_p \Im(\vec{i}_s \vec{\psi}_s)$$

arbitrary angular speed ω_{rf} 🙌

- slip: $s = 1 - \frac{\omega_r}{\omega_s} = 1 - \frac{n_p \omega_m}{\omega_s}$

Space Vector Orientation

Simulink Implementation

Considering the Park and Clarke transformations:

① Space Vector in Stationary Reference Frame

- $(\alpha\beta) \rightarrow \varphi = 0, \omega_{rf} = 0$
 $\vec{x}_{\alpha\beta} = \sqrt{\frac{2}{3}} (x_a + ax_b + a^2 x_c) = x_\alpha + jx_\beta \quad a = e^{j\frac{2\pi}{3}} \quad a^2 = e^{j\frac{4\pi}{3}}$

② Space Vector in Rotating Reference Frame

- $(dq) \rightarrow \varphi$ varying (we can select any rotating speed)
 $\vec{x} = \vec{x}_{\alpha\beta} e^{j\varphi} = \left[\sqrt{\frac{2}{3}} (x_a + ax_b + a^2 x_c) \right] e^{j\varphi} = (x_\alpha + jx_\beta) e^{j\varphi} = x_d + jx_q$

the angle is $\varphi = \int \omega_{rf} dt$ while $\vec{x} = \vec{v}_s, \vec{v}_r, \vec{i}_s, \vec{i}_r, \vec{\psi}_s, \vec{\psi}_r$

Model in Matrix Notation

Induction Motor

- The dynamic model of a three-phase IM with *orthogonal* axes dq is:

IM motor model (matrix form)

$$\mathbf{v}_s = R_s \mathbf{i}_s + p \boldsymbol{\psi}_s + \omega_{rf} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \boldsymbol{\psi}_s$$

$$\mathbf{v}_r = R_r \mathbf{i}_r + p \boldsymbol{\psi}_r + (\omega_{rf} - \omega_r) \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \boldsymbol{\psi}_r$$

$$\boldsymbol{\psi}_r = L_{ks} \mathbf{i}_s + \boldsymbol{\psi}_r$$

$$\boldsymbol{\psi}_s = L_m (\mathbf{i}_s + \mathbf{i}_r)$$

$$m_e = n_p (\boldsymbol{\psi}_s \times \mathbf{i}_s) = n_p (\psi_{sd} i_{sq} - \psi_{sq} i_{sd})$$

- remember that for a general variables $\mathbf{x} = \begin{bmatrix} x_d \\ x_q \end{bmatrix} = \begin{bmatrix} \Re(\vec{x}) \\ \Im(\vec{x}) \end{bmatrix}$
- it is valid for $\mathbf{x} = \mathbf{v}_s, \mathbf{v}_r, \mathbf{i}_s, \mathbf{i}_r, \boldsymbol{\psi}_s, \boldsymbol{\psi}_r \rightarrow \mathbf{v}_s = \begin{bmatrix} v_{sd} \\ v_{sq} \end{bmatrix} = \begin{bmatrix} \Re(\vec{v}_s) \\ \Im(\vec{v}_s) \end{bmatrix}$

Simulink Model in Space Vector Notation

Stationary alfa-beta model

- **Electrical part:** (squirrel cage $\rightarrow \vec{v}_r = 0$)

Stationary $\alpha\beta$ model

$$\vec{v}_{s\alpha\beta} = R_s \vec{i}_{s\alpha\beta} + p \vec{\psi}_{s\alpha\beta}$$

$$0 = R_r \vec{i}_{r\alpha\beta} + p \vec{\psi}_{r\alpha\beta} - j \omega_r \vec{\psi}_{r\alpha\beta}$$

$$\vec{\psi}_{s\alpha\beta} = L_{ks} \vec{i}_{s\alpha\beta} + \vec{\psi}_{r\alpha\beta}$$

$$\vec{\psi}_{r\alpha\beta} = L_m (\vec{i}_{s\alpha\beta} + \vec{i}_{r\alpha\beta})$$

$$m_e = n_p \Im \left(\vec{i}_{s\alpha\beta} \underline{\psi}_{s\alpha\beta} \right)$$

Classical form

$$\vec{\psi}_{s\alpha\beta} = \int (\vec{v}_{s\alpha\beta} - R_s \vec{i}_{s\alpha\beta}) dt$$

$$\vec{\psi}_{r\alpha\beta} = \int (-R_r \vec{i}_{r\alpha\beta} + j \omega_r \vec{\psi}_{r\alpha\beta}) dt$$

$$\vec{i}_{s\alpha\beta} = \frac{1}{L_{ks}} (\vec{\psi}_{s\alpha\beta} - \vec{\psi}_{r\alpha\beta})$$

$$\vec{i}_{r\alpha\beta} = \frac{\vec{\psi}_{r\alpha\beta}}{L_m} - \vec{i}_{s\alpha\beta}$$

$$m_e = n_p \Im \left(\vec{i}_{s\alpha\beta} \underline{\psi}_{s\alpha\beta} \right)$$

- **Mechanical part:**

$$J \frac{d\omega_m}{dt} + \beta \omega_m = m_e - m_l \quad \text{where} \quad \omega_r = n_p \omega_m$$

- Remember that j can be rewritten as $e^{j\frac{\pi}{2}} = \cos\frac{\pi}{2} + j\sin\frac{\pi}{2} = j$

Simulink Model in Space Vector Notation

Stationary alfa-beta model

Classical form (Complete)

$$\vec{\Psi}_{s\alpha\beta} = \int (\vec{v}_{s\alpha\beta} - R_s \vec{i}_{s\alpha\beta}) dt$$

$$\vec{\Psi}_{r\alpha\beta} = \int (-R_r \vec{i}_{r\alpha\beta} + j\omega_r \vec{\Psi}_{r\alpha\beta}) dt$$

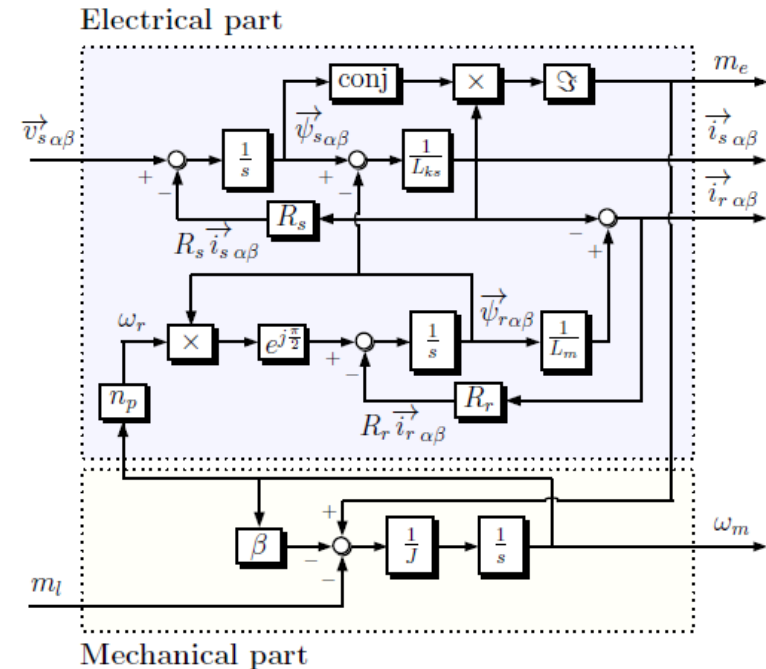
$$\vec{i}_{s\alpha\beta} = \frac{1}{L_{ks}} (\vec{\Psi}_{s\alpha\beta} - \vec{\Psi}_{r\alpha\beta})$$

$$\vec{i}_{r\alpha\beta} = \frac{\vec{\Psi}_{r\alpha\beta}}{L_m} - \vec{i}_{s\alpha\beta}$$

$$m_e = n_p \Im (\vec{i}_{s\alpha\beta} \vec{\Psi}_{s\alpha\beta})$$

$$\omega_m = \int \frac{1}{J} (m_e - m_l - \beta \omega_m) dt$$

- input: $\vec{v}_{s\alpha\beta}, m_l$
- output: $\vec{i}_{s\alpha\beta}, \vec{i}_{r\alpha\beta}, m_e$



→ Γ-model (space vector)

Brief Exercise

Simulink Model

Built a Simulink scheme of the previous IM model:

- create a subsystem for the space vector transformation
- create a IM subsystem in stationary $\alpha\beta$ components in sv form
- create a mask for input parameters

Hint:

- realize the following blocks:

